QUPON

A Project Based Learning Report Submitted in partial fulfilment of the requirements for the award of the degree

of

Bachelor of Technology

in The Department of CSE

FULL STACK APPLICATION DEVELOPMENT - 24SDCS02A

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FEB - 2025.

Introduction

In the current digital era, online shopping platforms and discount coupons have become widely popular as they help users save money. However, a large number of coupons remain unused and eventually expire, resulting in a loss of potential savings. There is no proper platform available where users can resell or reuse these unused coupons efficiently. This creates a gap in the market where both sellers and buyers miss valuable opportunities.

To solve this problem, we developed Qupon, a full stack web application designed to create a marketplace for unused coupons. The platform allows users to upload their unused coupons and sell them to interested buyers. Each coupon is verified through an admin system to ensure authenticity and prevent fraud. Buyers can browse available coupons, select suitable deals, and complete transactions securely using UPI.

The system is built using modern web technologies such as React and Next.js for the frontend, Node.js and Express for the backend, and MongoDB for data storage. The application ensures smooth interaction between users with a user-friendly interface and efficient backend processing.

Qupon not only reduces wastage of coupons but also promotes smart spending habits. It provides a secure, scalable, and efficient solution that can be expanded into a large-scale marketplace in the future.

Literature Review/ Application Survey

The rapid growth of e-commerce platforms has significantly increased the use of digital coupons and promotional offers. Many existing platforms such as Honey, RetailMeNot, and GrabOn provide users with discount coupons to enhance their shopping experience. These platforms primarily focus on distributing coupons offered by companies and businesses. However, they do not provide a system for users to resell or reuse unused coupons, which leads to wastage of valuable resources. Most traditional coupon systems operate in a one-way model where businesses provide discounts to consumers. Once a user obtains a coupon, it either gets used or expires without any alternative option. This limitation highlights the lack of a secondary marketplace where unused coupons can be exchanged or sold.

Qupon addresses this gap by introducing a peer-to-peer coupon resale system. Unlike existing platforms, it allows users to act as both buyers and sellers. This creates a dynamic marketplace where unused coupons gain value instead of being wasted. One of the major challenges in such systems is ensuring the authenticity of coupons. Qupon solves this problem by implementing an admin verification mechanism, where each uploaded coupon is reviewed before being made available for purchase.
Another key issue in coupon exchange systems is trust between users. Many platforms lack secure payment systems, which creates hesitation among users. Qupon integrates UPI-based payment solutions, which are widely trusted and used in India. This ensures safe and instant transactions between buyers and sellers.
From a technical perspective, modern full stack development frameworks have made it possible to build scalable and efficient applications. The MERN stack (MongoDB, Express, React, Node.js) is widely used due to its flexibility and performance.
Qupon uses this stack to ensure fast response times, efficient data handling, and a seamless user experience.

User engagement is another important factor in application design. Qupon incorporates features such as profile systems, level-based rewards, and notifications to keep users actively involved. These features are inspired by modern applications that use gamification techniques to improve user retention and satisfaction.

Scalability is also considered in the system design. With the help of cloud platforms like AWS and Vercel, the application can handle increasing user traffic and data efficiently. The use of MongoDB Atlas further enhances database performance with features like automatic backups and distributed storage.
The system also ensures smooth integration between frontend and backend components using RESTful APIs. This allows seamless communication and efficient data transfer between different parts of the application.

In conclusion, existing coupon platforms focus mainly on distribution, while Qupon introduces a unique approach by enabling resale and reuse. By combining secure transactions, verification systems, and modern technologies, Qupon provides a reliable and scalable solution. It stands out as an innovative platform that transforms unused coupons into valuable assets, benefiting both buyers and sellers.

References

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